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Exercise: Day 4

In this problem, you will evaluate the committer for a stochastic system on a bistable potential $w(q)$. The stable states (e.g., reactant and product) of this potential have values of q_A and q_B. Here, we will calculate the committor function (i.e., the splitting probability), $\phi(q)$, the fraction of trajectories initiated with a specific value of q that reach reach state B prior to reaching state A. To begin, consider the Smoluchowski equation,

$$\frac{\partial p(q,t)}{\partial t} = -\frac{\partial}{\partial q} J(q,t),$$

Where

$$J(q,t) = -De^{-\beta w(q)} \frac{\partial}{\partial q} \left(e^{\beta w(q)} p(q,t) \right)$$

1. The committor, $\phi_B(q_0)$, can be written in terms of the solution to the Smoluchowski equation,

$$\phi_B(q_0) = \int_0^\infty dt J(q_B, t)$$

with the following boundary conditions,

$p(q,0) = \delta(q - q_0)$ $\quad$ $p(q_A, t) = 0$ $\quad$ $p(q_B, t) = 0$

Provide a physical explanation for this expression and for each boundary condition.

2. Integrate the Smoluchowski equation from $t = 0$ to $t = \infty$. Simplify your result using the initial condition, $p(q,0) = \delta(q - q_0)$ and the fact that the boundary conditions steadily remove probability from the interval $q_A < q < q_B$.

3. Integrate your result from part 2 over the coordinate q, Take q_B as the upper limit of this integration. For the lower limit, use an arbitrary value of q. In your result, you should be able to identify the committor, $\phi_B(q_0)$

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4. Multiply your result from part 3 by the inverse Boltzmann factor, $\exp(\beta w(q))$ and integrate once again over q. This time, take q_A and q_B as the limits of integration. A simple rearrangement should yield Onsager's result:

$$\phi_B(q_0) = \frac{\int_{q_A}^{q_0} dq e^{\beta w(q)}}{\int_{q_A}^{q_B} dq e^{\beta w(q)}}$$

5. Consider the idealized case of a harmonic barrier:

$$w(q) = -\frac{1}{2} m \omega^2 q^2$$

Setting $q_A = -\infty$ and $q_B = \infty$, evaluate the splitting probability using the equation above.
Your answer should involve the error function,

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x dz e^{-z^2}$$

6. For the harmonic barrier model, plot $\phi_B(q_0)$ as a function of q_0 over a range that encompasses most of the crossover between its limiting behaviors. Estimate the width of this crossover, i.e., a range of q_0 over which $\phi_B(q_0)$ varies between 0.1 and 0.9.